Data Association for Multiple Hypothesis Tracking on a Neutral-Atom Quantum Computer

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21/21 instances solved exactly

19.2 benchmark reproduced

0.86× sampling gain (<1 = none)

34% shots violating a constraint

12 / 256 atoms needed / available

1. The problem

A sensor returns, at every time step, a set of **unlabelled** detections. Nothing says which object produced which detection, some objects are missed, and some detections are spurious clutter.

Deciding which detections belong together is **data association**. Committing early is dangerous: if two objects cross and their identities swap, the error propagates forever.

Multiple Hypothesis Tracking (MHT) avoids this by keeping many competing explanations alive and deferring the decision. The price is combinatorial growth — and that is where a quantum computer was proposed to help.

2. Tracking → graph problem

Following Papageorgiou & Salpukas [1], each candidate **track** becomes a vertex weighted by its log-likelihood-ratio score, and an edge joins two tracks that claim the **same detection** — one detection cannot come from two objects.

The best global hypothesis is then a **Maximum Weight Independent Set (MWIS)**:

$$\max \sum_i w_i x_i \quad \text{s.t.} \quad x_i + x_j \le 1 \;\; \forall (i,j) \in E.$$

The weight is built recursively from the sensor model,

$$\Delta\mathrm{LLR} = \begin{cases} \log(1-P_D) & \text{missed}, \\ \log \dfrac{P_D \mathcal{N}(\nu; 0, S)}{\beta_{\mathrm{FA}}} & \text{detected}, \end{cases}$$

where β_{FA} is the clutter density: in a clean scene a coincidence means something, in a noisy one it does not.

MHT needs more than the best answer. Track confidence requires a *population* of hypotheses:

$$P\{H_j\} = \frac{e^{L_{H_j}}}{1 + \sum_i e^{L_{H_i}}}, \qquad P\{T_i\} = \sum_{j: T_i \in H_j} P\{H_j\}.$$

This is the property the quantum approach was meant to exploit.

3. Why neutral atoms

Two atoms closer than the **blockade radius**

$$R_b = \left(\frac{C_6}{\hbar\Omega}\right)^{1/6}$$

cannot both be excited to a Rydberg state. Placing incompatible tracks within R_b enforces $x_i + x_j \le 1$ **in hardware**, with no penalty term to tune:

$$H/\hbar = -\sum_i \delta_i n_i + \sum_{i<j} \frac{C_6}{r_{ij}^6} n_i n_j.$$

Track scores become **local detunings** via the Detuning Map Modulator. It only shifts downwards, so $m_i = 1 - w_i/w_{\max}$ gives $\delta_i = \delta_{\max} w_i / w_{\max}$ — needing $w_i > 0$, which the "positive-score tracks only" rule supplies for free.

4. Pipeline

Classical: Kalman prediction → gating → track branching → LLR scoring → pruning, merging, N-scan pruning → clustering into independent sub-problems.

Quantum, per cluster: geometric embedding → optical-tweezer register → adiabatic sweep → measure → repeat.

Classical pruning is what makes it fit. The largest cluster over a whole scenario was **9 vertices** — far inside a 256-atom device. Gating, confirmation, track merging and N-scan pruning, *not* the quantum step, keep the problem small. Drop track merging and clusters inflate into near-cliques no planar register can hold.

5. The geometric embedding

On hardware you do **not** choose the edges — physics decides them from distance. The graph solved is always a **unit-disk graph**; an MHT graph is not. We need

$$\|p_i - p_j\| < R_b \text{ (edge)}, \qquad \|p_i - p_j\| > R_b \text{ (non-edge)},$$

solved by penalty minimisation with random restarts, in units of R_b.

Ω is derived, not chosen. The layout fixes only distance *ratios*. A larger R_b gives atoms room but costs Rabi frequency as R_b^{-6}, and too small an Ω never leaves the ground state inside the device's 6 μs limit.

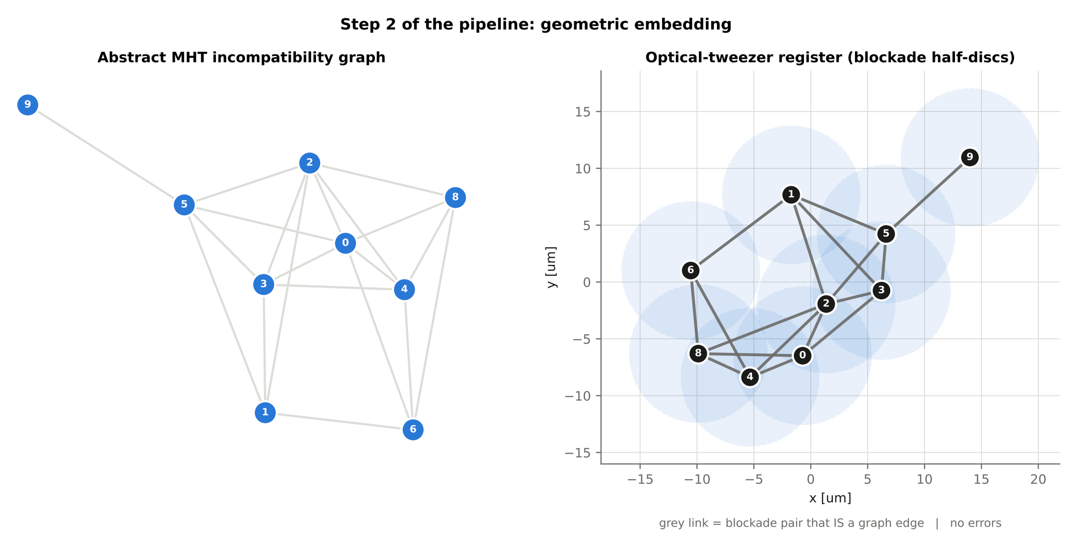

Figure 1: A real MWIS instance from the tracker: abstract conflict graph (left) and the atom register realising it (right). Shaded discs show blockade range.

Cliques are the obstruction. A k-clique needs k atoms pairwise *within* R_b yet pairwise $\ge 5\,\mu$m apart. Feasibility depends only on $\rho = R_b/d_{\min} \approx 2.4$: a hexagon plus centre gives 7 points at ratio exactly 2, so ≥ 7 fit; the limit is ≈ 9. Each track family *is* a clique — so capping branches per family is a hardware constraint, not a tuning choice.

6. Validation

We reproduce the worked example of [1] — six tracks with published incompatibility lists; optimum $\{3, 6\}$, weight **19.2**. Classical solver and atom array both return exactly this, the array sampling it in **93–95%** of shots with a defect-free embedding.

In closed-loop tracking on synthetic radar (5 converging targets, $P_D = 0.9$, Poisson clutter) the array solved **21 instances and matched the exact optimum on all 21**, following every object with 100% association accuracy.

7. Testing the advantage

Optimisation alone proves nothing here — classical solvers find the optimum instantly at $n \le 12$. The real claim concerns $P\{H_j\}$: the array should give a *distribution* at fixed cost, while classical enumeration cost grows with how many hypotheses exist (up to $3^{n/3}$).

Real data. `PhC-C2DL-PSC`, Cell Tracking Challenge [2]: 300 frames of phase-contrast microscopy, cells proliferating $74 \to 661$. Shared deterministic detections (six-frame window: recall 0.64, precision 0.86, centroid F1 0.73 at 10 px); human tracking ground truth grades the detector only. Six MWIS instances of 9–12 tracks.

Metric: posterior mass per sample. Counting hypotheses would rate a 0.1% explanation equal to one carrying 90%. A "sample" = one measure cycle / annealing run / construction; identical post-processing for all.

Sampler	Samples	Prec.
Simulated annealing	**23.6**	98.0%
Neutral atoms	27.5	97.8%
Randomised greedy	35.3	91.0%
Atoms, no repair	32.2	64.2%

Table 1: Samples to capture 90% of the posterior mass.

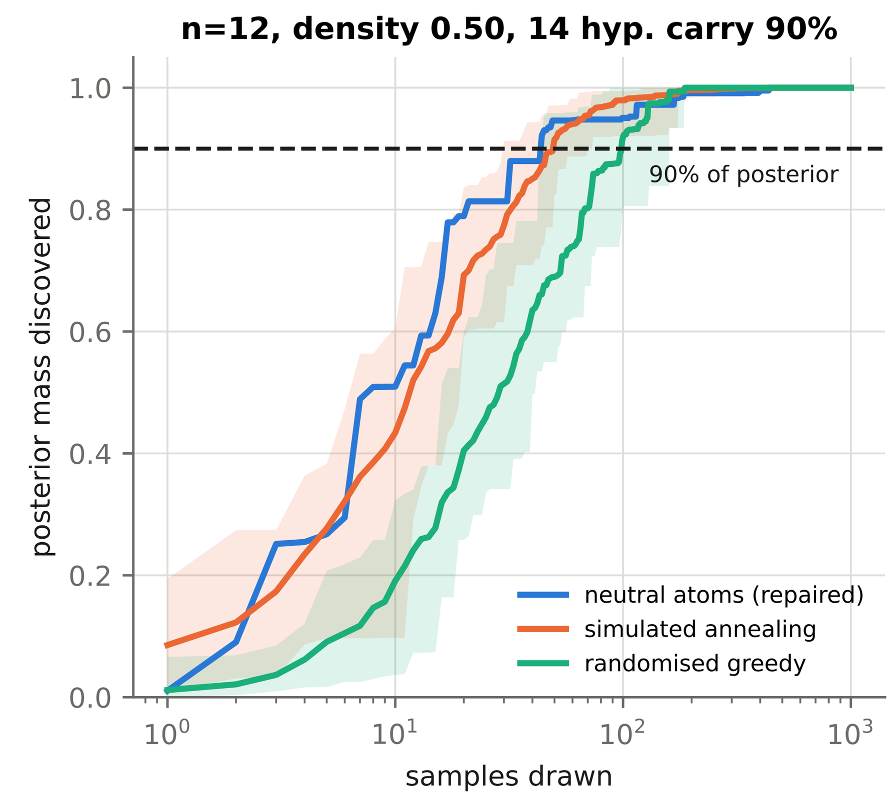

Figure 2: Posterior mass discovered against samples drawn, for one instance. Shading spans repeats of the classical samplers.

The claim is not supported. The array needed *more* samples than annealing. The margin is small — it won on 4 of 6 instances and beat randomised greedy clearly — so the honest reading is "indistinguishable at this sample size", not a clear loss. Six instances cannot settle it either way.

8. Why

Embeddability dominates. Real conflict graphs had mean density **0.66**, and **34%** of shots violated an edge. The blockade — the one mechanism meant to enforce constraints for free — was not enforcing them. Classical repair then does that work, and a repaired shot is no longer a physical sample.

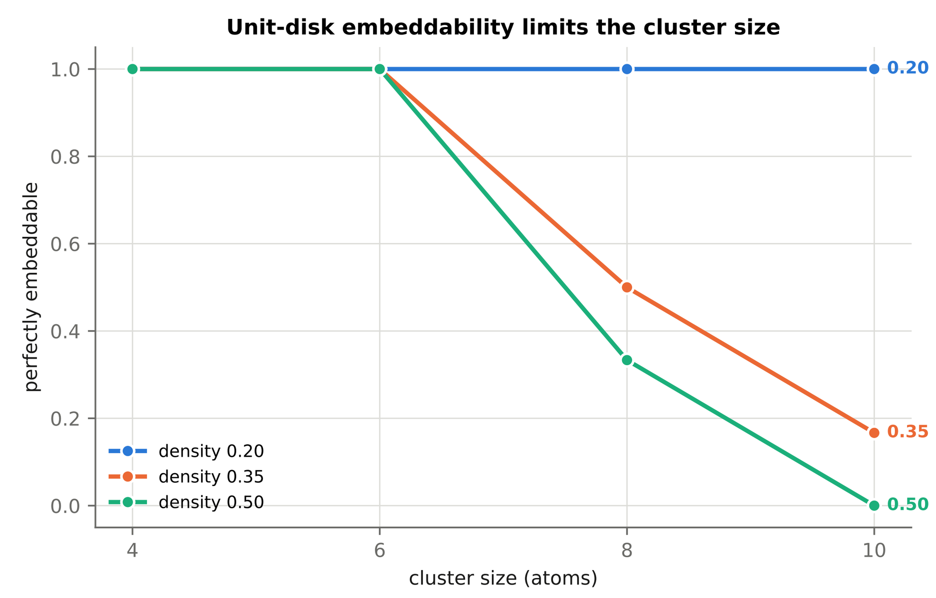

Figure 3: Perfect unit-disk embeddings vanish with size and density. The limit is geometry, not qubit count.

Concentration fights coverage. Capturing 90% of the posterior needs ≈ 7 distinct hypotheses, but an adiabatic sweep returns *one*. Sweep duration moves $P(\text{optimum})$ from 8% to 99% while diversity moves the other way — a knob with no classical analogue, but not enough to close the gap.

9. Conclusions

- The physics mapping is sound; the implementation reproduces published results exactly.
- **λ is not free.** The penalty is C_6/r_{ij}^6, set by atom placement. You choose $\delta_{\max}/\Omega$, bounded on *both* sides; measured window $2 \lesssim \delta_{\max}/\Omega \lesssim 5.6$.
- **Geometry binds, not qubit count.** We never needed more than 12 of 256 atoms.
- **Weights are quantised:** score differences below ≈ 2.6 LLR are invisible to the hardware.

Consequence. Making the graph *natively* unit-disk is a **precondition**, not a refinement. Untested and promising: place atoms by proximity in *track state space*, where tracks sharing a detection are already close.

Open: repair bias; unresolved targets (two objects merging into one detection *should* share it); hardware noise, which hits the weight encoding directly since the weights *are* detunings.

Refs. [1] Papageorgiou & Salpukas, Springer 2009, 235–255. [2] Ulman et al., *Nat. Methods* **14**, 1141 (2017), celltrackingchallenge.net. [3] Silvério et al., *Quantum* **6**, 629 (2022). [4] Ebadi et al., *Science* **376**, 1209 (2022). **Code:** `code/`, Pulser/QuTiP emulator.